

# Performance Metrics in Networks

Validate Protocol

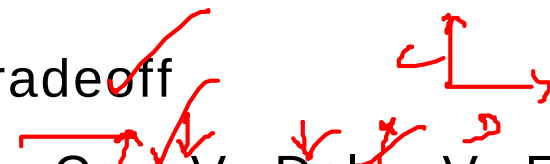
Dr. N.Sujaudeen  
Assistant Professor, Dept. of CSE

This presentation covers

- Throughput
- Latency / Delay
- Loss



# Challenges

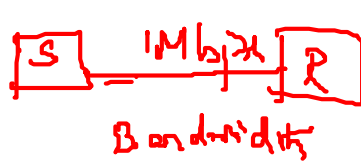
- Tradeoff 
  - Cost Vs Delay Vs Energy
  - Need to strike the right balance based on usage scenario.
- System Failure 
  - Nodes can die, links corrupt packets, processing can duplicate, reorder packets.
- Backward Compatibility 
  - Newer version of protocols should support older devices.

Throughput  $\rightarrow$  bandwidth  
 Mbps, Kbps  
 Megabits

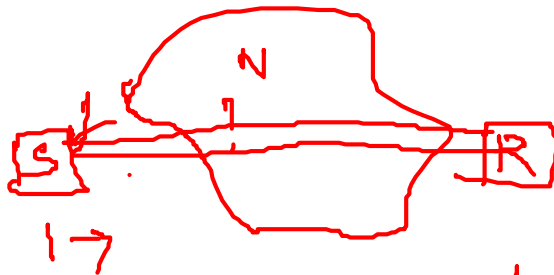
$-(Hz) \rightarrow$  Spectrum  
 MBps / Kbps

$$\text{Receiving Throughput} = \frac{\text{Amount of data}}{\text{time it took to receive the data.}}$$

$$= \frac{10^6 \text{ bits}}{1} = 1 \text{ Mbps}$$

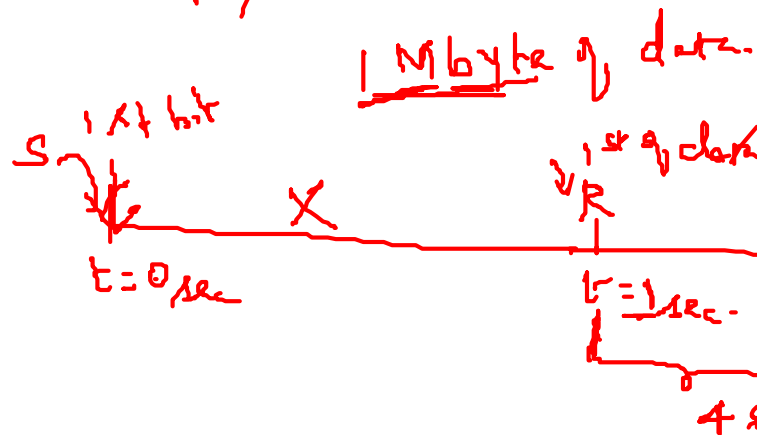


$10^6 \text{ bits} / 1 \text{ sec.}$



$$\text{Transfer Throughput} = \frac{\text{Amount of data}}{\text{Time to send data from S to R. (Time when last bit is received)}}$$

$$= \frac{10^6 \times 1.6}{8} = 1.6 \text{ Mbps}$$



$$\text{Last bit of data} = \frac{10^6 \times 8}{2} = 4 \text{ Mbps}$$

$$= \frac{10^6 \times 2}{2} = 1 \text{ Mbps}$$

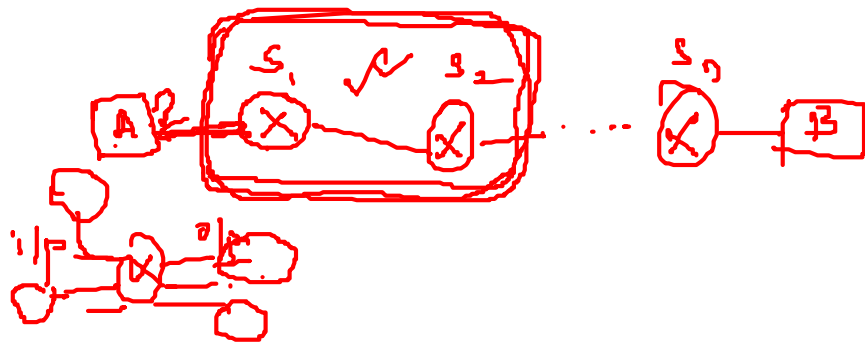


Latency  
Delay → time taken by the packet.  
↳ S → D. (1 way delay)

RTT - Round trip time

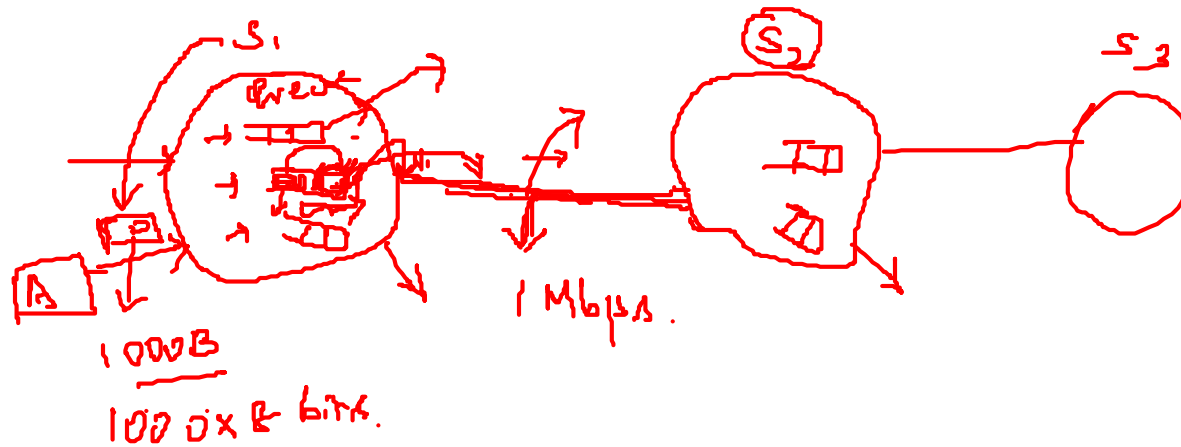
- MS, ms, s.

↳ processing delay,  
- Transmission  
- Queuing  
- Propagation.



Processing delay  
 ↳ Examine the packet.

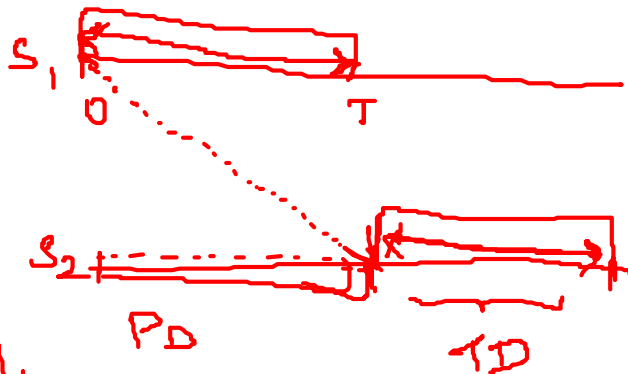
Queueing delay = Waiting in the queue.



$$\text{Transmission delay} = \frac{\text{Size}}{\text{band width}}$$

$$= \frac{1000 \times 8}{10^6} = 8 \text{ ms}$$

Propagation delay =  $\frac{\text{distance}}{\text{speed of light}}$



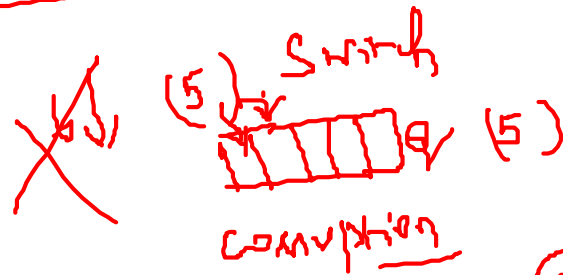
Delay / Latency = PD + QD + TD + Prop D.

Cable -  $2.3 \times 10^8 \text{ m/s}$   
 fibre -  $2 \times 10^8 \text{ m/s}$

Vacuum =  $3 \times 10^8 \text{ m/s}$



Loss:



→ probability. 0-1

$$\frac{10\%}{100} =$$

$$\frac{1 \times 10}{10 \times 10}$$

0.1

0.1

10% of every 10 packets  
Average of 1

Parameter → induce the loss.

or  
metric.  
&  
measure.

↓  
parameter work

Thank You...

